Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2025-P2-Model)** **Date: 05-04-2026**

Time: 02.00Pm to 05.00Pm **GTA-22** **Max. Marks: 180**

KEY SHEET

MATHEMATICS

1	**C**	2	**C**	3	**A**	4	**D**	5	**C**	6	**AD**
7	**ABCD**	8	**BCD**	9	**0**	10	**184**	11	**105**	12	**28**
13	**170**	14	**117**	15	**90**	16	**8**				

PHYSICS

17	**C**	18	**B**	19	**B**	20	**C**	21	**ABCD**	22	**AC**
23	**ABCD**	24	**ABC**	25	**0.25**	26	**1.57**	27	**56.56 to 56.57**	28	**2.25**
29	**1.3**	30	**7.20**	31	**24.62**	32	**7**				

CHEMISTRY

33	**C**	34	**B**	35	**D**	36	**D**	37	**BC**	38	**ACD**
39	**ABC**	40	**BD**	41	**9**	42	**10**	43	**5**	44	**10**
45	**319**	46	**8**	47	**4**	48	**5**				

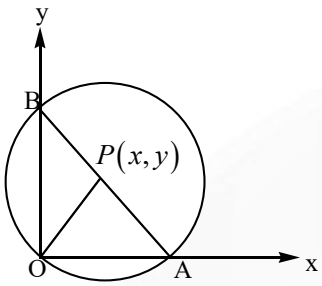
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SOLUTIONS

MATHEMATICS

1. Image of Z in the line $y=x$ is $i\bar{z}$

2.



Equation of AB is $y-y_1=-\frac{x_1}{y_1}(x-x_1)$ $\quad OA^2+OB^2=(2r)^2$

3. $\sum_{r=1}^{10}\sin\left(3^{r-1}\theta\right).\sec\left(3^r\theta\right)=\frac{1}{2}\sum_{r=1}^{10}\frac{\sin\left(3^r\theta-3^{r-1}\theta\right)}{\cos\left(3^{r-1}\theta\right)\cos\left(3^r\theta\right)}$

$=\frac{1}{2}\left[\tan\left(3^{10}\theta\right)-\tan\theta\right]=-\frac{1}{2}\tan\left(\frac{\pi}{3^5}\right)$

4. $y=\left(x+\sqrt{x^2-1}\right)^5$

Difference in twice w.r to x $\quad \left(x^2-1\right)\frac{d^2y}{dx^2}+x\frac{dy}{dx}=25y$

5. $\int\frac{\tan^{-1}x}{x^4}dx=\int\frac{\tan^{-1}x}{4f}\frac{1}{x^4}dx$

Apply by parts

$\int\frac{\tan^{-1}x}{x^4}dx=-\frac{1}{3x^3}\left(\tan^{-1}x\right)+\frac{1}{6}\log\left|\frac{x^2}{x^2+1}\right|-\frac{1}{6x^2}+c$

6.

7. $P^8+P^6=18I-29P \quad P^8-P^6=8I-13P$

8. Shortest distance $=2\sqrt{6}$

$|d+3|=12,\ \Rightarrow h=9,-15$

But $h=9$

9. $f(x)=\begin{cases}8+2x;-4\le x<-2\\ x^2;-2\le x\le -1\\ |x|;-1<x<1\\ x^2;1\le x\le 2\\ 8-2x;2<x\le 4\end{cases}$

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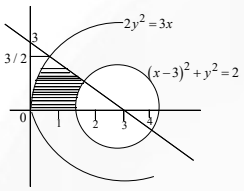
Clearly f(x) is not differentiable at $x=\{\pm 2,\pm 1,0\}$

10. $a+b+c+d=5 or 7 or 11,\ where\ A=\begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$$\left(1+x+x^2+x^3+x^4\right)^4=\left(1-x^5\right)^4(1-x)^4$$

Sum of the coefficients of $x^5, x^7\ and\ x^{11}$ in the expansion of $\left(1-x^5\right)^4(1-x)^{-4}$

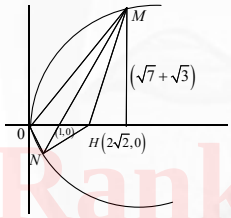
11.



$$A=\int_0^{3/2}\left(3-y-\frac{2y^2}{3}\right)dy-\frac{2\pi}{8}\Rightarrow 4A+\pi=\frac{21}{2}$$

12. $y=mx+c$ passes through the focus (1,0)

$m+c=0$



$y=mx-m$ is a tangent to hyperbola $x^2-y^2=4$ $\quad m=\frac{2}{\sqrt{3}}$

So focal chord $y=\frac{2}{\sqrt{3}}x-\frac{2}{\sqrt{3}}$

Solving this with parabola $y=\sqrt{3}\pm\sqrt{7}$

H=Focus of the hyperbola $=\left(2\sqrt{2},0\right)$

13. $x=x+h\ and\ y=y+x$

$$\tan^{-1}\left(\frac{y-1}{x-1}\right)-\frac{1}{2}\ln\left(1+\left(\frac{y-1}{x-1}\right)^2\right)=\ln|x-1|$$

$\alpha=10\ and\ \beta=20$

14. let the plane is $L_1+\lambda L_2=0$

15. From partial fractions

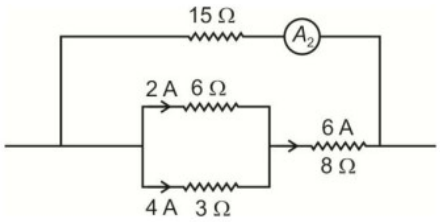
$$A_{13}=\frac{-1}{136.76}, A_{14}=\frac{1}{146.66}, A_{15}=\frac{-1}{156.56}$$

16. Apply L-hospital rule $\Rightarrow L=\frac{1}{9}$

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PHYSICS

17. Use of Kirchhoff's law



$15 \times l_2 = 6 \times 2 + 6 \times 8 \qquad \Rightarrow l_2 = 4A$

18. $\dfrac{dV}{dt} = \dfrac{F}{m} \qquad -mg(dr) - Fv\,dt = \dfrac{d}{dt}\left(\dfrac{1}{2}mv^2\right)$

$\Rightarrow -mg\dfrac{dr}{dt} = FV + mv\left(\dfrac{dV}{dt}\right) = Fv + Fv \qquad \Rightarrow -\dfrac{dr}{dt} = \dfrac{2FV}{mg}$

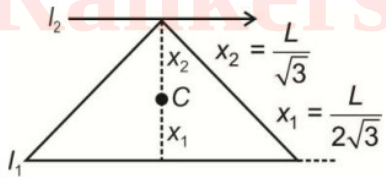
19. $V = I_{ms}X_C \qquad 220 = \dfrac{220}{Z}X_C$

$\Rightarrow X_C = Z = |X_C - X_L| \qquad \Rightarrow X_C = X_L - X_C$

$\Rightarrow X_C = \dfrac{X_L}{2} = \dfrac{\omega L}{2} \qquad \Rightarrow \dfrac{1}{\omega C} = \dfrac{\omega L}{2}$

$\Rightarrow C = \dfrac{2}{\omega^2 L} = \dfrac{2}{(100\pi)^2 \times 1} \qquad C = 20\mu F$

20. $I = \int (dm)r^2$



$l_1 = \dfrac{Mh^2}{6} = \dfrac{M}{6} \times \left(\dfrac{\sqrt{3}}{2}L\right)^2 = \dfrac{ML^2}{8}$

$l_1 = I_c + Mx_1^2 \qquad l_2 = I_c + Mx_2^2 = l_1 - mx_1^2 + Mx_2^2 = \dfrac{ML^2}{8} + M\left(\dfrac{L^2}{4}\right) = \dfrac{3}{8}ML^2$

21. $I = \dfrac{\varepsilon}{\frac{2R}{3}} = \dfrac{3\varepsilon}{2R}$

$= \dfrac{3}{2R} \times \dfrac{1}{2}B\omega\ell^2 = \dfrac{3B\omega\ell^2}{4R}$

Magnetic force $F = \dfrac{3B\omega\ell^2}{4R} \times \ell \times B$

$= \dfrac{3B^2\omega\ell^2}{4R}$

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$$\tau = \frac{3B^2\omega\ell^3}{4R}\times\frac{\ell}{2} = \frac{3B\omega\ell^4}{8R}$$

$$\therefore \text{ force to be applied at the end } = \frac{3B^2\omega\ell^3}{8R}$$

22. $\Delta x = 3\sin\theta = \lambda \quad \Rightarrow \sin\theta = \frac{1}{3} \Rightarrow \tan\theta = \frac{1}{\sqrt{8}}$

$$\therefore y_1 = 1\times\frac{1}{\sqrt{8}} = \frac{1}{\sqrt{8}}m \text{ and}, 3\sin\theta = 2\lambda \Rightarrow \sin\theta = \frac{2}{3} \Rightarrow \tan\theta = \frac{2}{\sqrt{5}} \therefore y_2 = \frac{2}{\sqrt{5}}m$$

23. $kx_1 = F + (3m)g \quad kx_2 = (2m)g$

$$\text{and } \frac{1}{2}kx_1^2 = \frac{1}{2}kx_2^2 + (3m)g(x_1 + x_2) \quad \Rightarrow F = (5m)g$$

24. Potential on innermost shell is zero

$$\frac{q_1}{r} + \frac{q_2}{2r} + \frac{q_3}{3r} = 0$$

$$\Rightarrow 6q_1 + 3q_2 + 2q_3 = 0 \text{ ..... (1)}$$

Potential on outermost shell is zero

$$\frac{q_1}{3r} + \frac{q_2}{3r} + \frac{q_3}{3r} = 0 \Rightarrow q_1 + q_3 = -q_2 \text{ ..... (2)}$$

25. For maximum current, shape is circular

$$\phi = \alpha t\times\pi\times\left(\frac{I}{2\pi}\right)^2, I\times A = \frac{m}{d} \qquad \Rightarrow I_m = \frac{\left(\frac{d\phi}{dt}\right)}{R} = \frac{\alpha m}{4\pi\rho d} \therefore k = 0.25$$

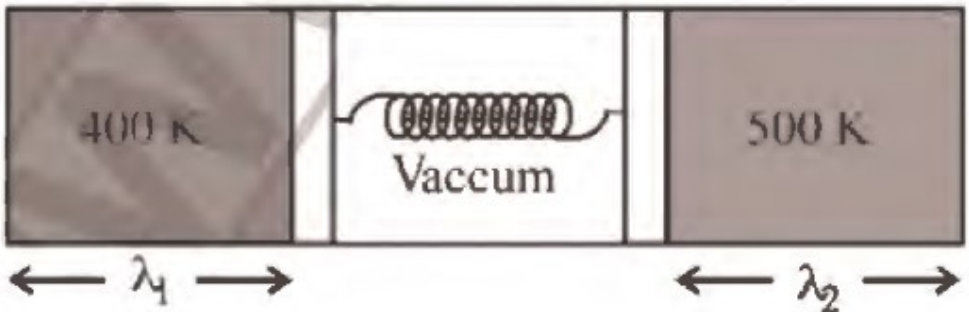
26. $ml^2\frac{d^2\theta}{dt^2} = -\left(k\theta\times\frac{I}{2}\sin\theta\right)\times\sin\theta\times\frac{I}{2} \Rightarrow T = \frac{2\pi\times 2}{\left(\frac{4}{5}\right)}\times\sqrt{\frac{1}{100}} = \frac{5\pi}{10} = \frac{\pi}{2} = 1.57s$

27. $y^2 = 2\times 2 \Rightarrow y = 2 \quad \therefore I_e = \sqrt{2^2+2^2} = 2\sqrt{2}m \qquad \therefore F_m = 5\times 2\sqrt{2}\times 4 = 56.57N$

28. $\Delta x = (odd\ number)\times\frac{\lambda}{2}$, For second minima, $\Delta x = 1.5\lambda$

$$\Rightarrow \sqrt{(3\lambda)^2 + x^2} - x = \frac{3\lambda}{2} \Rightarrow 9\lambda^2 + x^2 = x^2 + \frac{9\lambda^2}{4} + 3\lambda x \Rightarrow x = \frac{9}{4}\lambda = 2.25\lambda$$

29. Let l_1 *and* l_2 be the final lengths of the two parts, the from gas equation



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$$\frac{P_0 A\ell_0}{T_0}=\frac{PA\ell_1}{T_1}=\frac{PA\ell_2}{T_2} \quad \text{... (1)}$$

$P_0A=kx_0$ and $PA=kx$

$$\frac{P}{P_0}=\frac{x}{x_0} \quad \text{..... (2)}$$

$\therefore x-x_0=\ell_1+\ell_2-2\ell_0$ ..... (3)

For equation (1)

$$\ell_1=\frac{P_0\ell_0T_1}{PT_0} \text{ and } \ell_2=\frac{P_0\ell_0T_2}{PT_0}$$

For equation (2)

$$\ell_1=\frac{x_0\ell_0T_1}{xT_0} \text{ and } \ell_2=\frac{x_0\ell_0T_2}{xT_0}$$

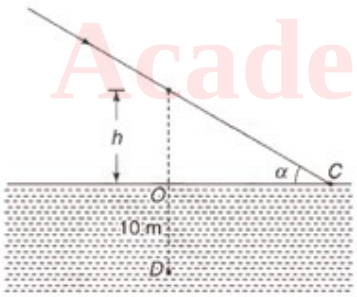
Putting these in equation (iii),

$$x-x_0=\frac{x_0\ell_0}{xT_0}[T_1+T_2]-2\ell_0$$

Substituting the values and solving for x, we get $x\approx 1.3m$

30. $$p=\frac{h}{\lambda}=hR\left(\frac{1}{2^2}-\frac{1}{3^2}\right)\Rightarrow mv_{recoil}=hR\times\left(\frac{5}{36}\right) \quad \Rightarrow V_{recoil}=\frac{1}{7.2}\left(\frac{hR}{m}\right)$$

31.



$h_{app}=\mu h$

$h_{app}=50-d=40m$ and $m=\mu=4/3$

$\therefore\ h=30m$

$Now, h_{app}=\mu h$

$V_{y\,app}=\mu V_y$

$V_x app=V_x$

$$Given \frac{V_{yapp}}{V_{xapp}}=\tan 45^o=1$$

$$\therefore \frac{\mu V_y}{V_x}=1\Rightarrow \tan\alpha=\frac{1}{\mu}=\frac{3}{4}$$

$\alpha=37^o$

α is the true angle that the line of motion of the bird makes with horizontal.

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$\therefore OC = h\tan a = 30.\frac{3}{4}$

$= 22.5m$

$\therefore DC = \sqrt{(22.5)^2 + 10^2} = 24.62m$

32. Least count is $\frac{1}{100} = 0.01mm$

Since circular scale lies 5 division below the reference line
So zero error is positive and its value is 0.05mm
Now on reading 3.40 mm-0.05 mm=3.35mm

$\therefore \frac{100}{5}(3.35-3) = \frac{35}{5} = 7$

CHEMISTRY

33. Cu, Hg, Pb, Ag

34. $\frac{N}{N_0} = e^{-E/2RT} = e^{-\frac{46\times10^3}{2\times2\times1000}} = 1.0\times10^{-5}$

In NaCl, all octahedral voids are occupied. Hence, the available voids are only tetrahedral
$\therefore N_i = 2\times N_0$

$Now,\ N = \sqrt{N_0\times 2N_0}.e^{-E/2RT}$

$\therefore \frac{N}{N_0} = \sqrt{2}\times e^{-E/2RT} = \sqrt{2}\times e^{-\frac{73.6\times1000}{2\times2\times1000}}$

$= 1.41\times10^{-8}$

35. i)

$$SO_3 \rightleftharpoons SO_2 + \frac{1}{2}O_2$$

No.of moles before dissociation 1 0 0

No.of moles after dissociation $1-\alpha$ α $\frac{\alpha}{2}$

Total moles $1-\alpha+\alpha+\frac{\alpha}{2} = 1+\frac{\alpha}{2}$

$$\frac{D}{d} = 1+\frac{\alpha}{2} \Rightarrow \alpha = \frac{2(D-d)}{d}$$

ii)

$$N_2O_4 \rightleftharpoons 2NO_2$$

No.of moles before dissociation 1 0

No.of moles after dissociation $1-\alpha$ 2α

Total moles $1-\alpha+2\alpha = 1+\alpha$

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$$\frac{D}{d}=1+\alpha \Rightarrow \alpha=\frac{(D-d)}{d}$$

iii)

$$PH_3 \rightleftharpoons \frac{1}{4}P_4+\frac{6}{4}H_2$$

No.of moles before dissociation 1 0 0

No.of moles after dissociation $1-\alpha$ $\frac{1}{4}\alpha$ $\frac{6}{4}\alpha$

Total moles $1-\alpha+\frac{1}{4}\alpha+\frac{6}{4}\alpha=1+\frac{3}{4}\alpha$

$$\frac{D}{d}=1+\frac{3\alpha}{4} \Rightarrow \alpha=\frac{4(D-d)}{3d}$$

iv)

$$A \rightleftharpoons \frac{6}{5}B+\frac{9}{5}C$$

No.of moles before dissociation 1 0 0

No.of moles after dissociation $1-\alpha$ $\frac{6}{5}\alpha$ $\frac{9}{5}\alpha$

Total moles $1-\alpha+3\alpha=1+2\alpha$

$$\frac{D}{d}=1+2\alpha \Rightarrow \alpha=\frac{(D-d)}{2d}$$

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36. Condition for 1,4 addition is conjugated should be in S-cis conformations.
A is present in S-cis conformations, 6-MCTS is possible so (1,4 addition). B is locked in S-trans conformations, only 4-MCTS is possible so 1,2 addition)

37. $Pb(NO_3)_2 \xrightarrow{\Delta} PbO+NO_2+O_2$
Same gases are formed with all Nitrates of alkaline earth metals, Nitrates of Li, Al. Zn, Cu, Hg etc. But alkali metal nitrates give metal Nitrite & O_2

38. Cupellation is used to remove the impurities of more electropositive metals in their oxide form present in had lead or impure lead. Ag is removed from lead by parke-pattinson process

39. Only $C_2O_4^{2-}$ ions are oxidised by $KMno_4$ solution
Normality of $KMno_4$ solution = 0.02 x 5 = 0.1 N

22.6 mL of 0.1 N $KMno_4$ = 22.6 mL of N $C_2O_4^{2-}$ soln.

Mass of $C_2O_4^{2-}$ ions = $\frac{N\times E\times V}{1000}=\frac{N\times M\times V}{1000\times 2}$

No.of moles of $C_2O_4^{2-}$ ions in the solution = $\frac{N\times M\times V}{1000\times 2\times M}$

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$$= \frac{N \times V}{2000}$$

$$= \frac{0.1 \times 22.6}{2000}$$

$$= 11.3 \times 10^{-4}$$

Second case: Only ions are reduced by KI and iodine liberated is neutralized by $Na_2S_2O_3$ solution

11.3 mL of 0.05 M $Na_2S_2O_3$ = 11.3 mL of 0.05 N $Na_2S_2O_3$

= 11.3 mL of 0.05 $N\ I_2$

= 11.3 mL of 0.05 $N\ Cu^{2+}$

Mass of Cu^{2+} ions in the solution $= \frac{N \times E \times V}{1000} = \frac{N \times M \times V}{1000 \times 2}$

No.of moles of Cu^{2+} ions in the solution $= \frac{N \times M \times V}{1000 \times M}$

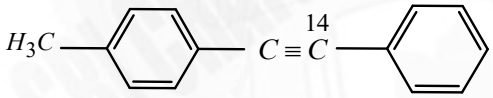
$$= \frac{N \times V}{1000}$$

$$= \frac{0.05 \times 11.3}{1000}$$

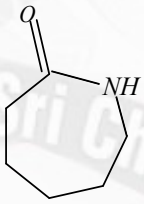
$$= 5.65 \times 10^{-4}$$

Molar ratio of $\frac{Cu^{2+}}{C_2O_4^{2-}} = \frac{5.65 \times 10^{-4}}{11.3 \times 10^{-4}} = \frac{1}{2}$

40. For B correct major product is



For D correct major product is



41.

$Na\left[Co(bipy)Cl_4\right]$

$1 + x + 0 - 4 = 0$

$1 + x = 4$

$x = 4 - 1 = +3$

Bipyridine bidentate ligand + chlorido

Monodentate ligand = 6

Sum = 3+6 =9

42.

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$_{67}HO-[Xe]4f^{"}6s^{2}$

$HO^{+3}-[Xe]4f^{10}6s^{0}$

43. B, F, H, I, L are correct statements

44.

$\Delta Q = cv\Delta T$

$= 2.5\times 0.45 = 1.125$

$\Delta E = \frac{28\times 1.125}{3.5} = 9kJ^{-1}mol^{-1}$

$2CO(g)+O_2(g)2CO_2(g)$

$\Delta H = \Delta E + \Delta n.RT$

$\Delta H = 2+\frac{1}{2}\times 8\times 10^{-3}\times 300$

$= 9+1.2 = 10.2\ kJ\ mol^{-1}$

45. The time taken for a reaction in inversely proportional to the rate constant k

$\frac{k_2}{k_1}=\frac{t_1}{t_2}\Rightarrow\frac{k_2}{k_1}=\frac{6}{1}=6$

The relationship between the rate constant at two temperatures is given by

$\frac{k_2}{k_1}=Q_{10}^{\frac{T_2-T_1}{10}}$

$6=3^{\frac{T_2-T_1}{10}}$

Taking natural log on both sides

$\ln(6)=\ln 3^{(T_2-30)/10}$

$\ln 6=\frac{T_2-30}{10}\times 1.099$

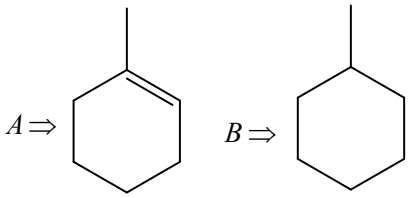
$1.791=\frac{T_2-30}{10}\times 1.099$

$\frac{1.791}{1.099}=\frac{T_2-30}{10}$

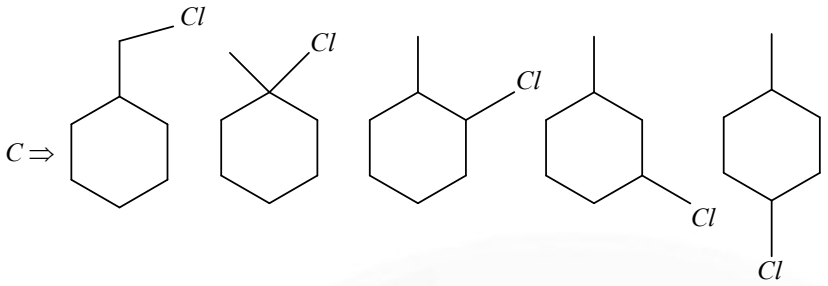
$16.29 = T_2-30$

$T_2 = 46.29^{0}C = 319.29\,K$

46.

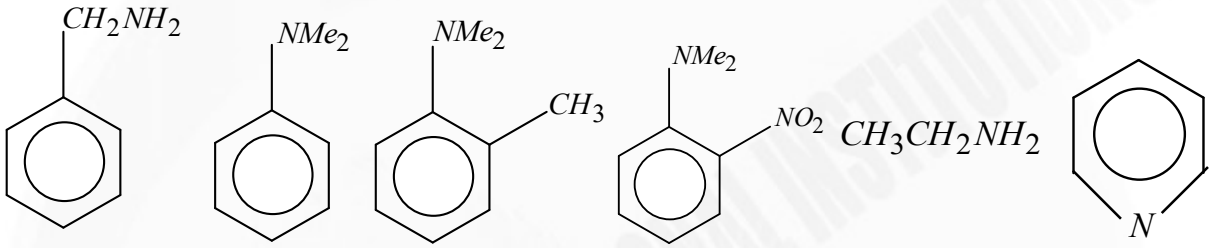


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47. $CH_3 - CH_2 - C \equiv C - Br,\ BrCH_2 - CH_2 - C \equiv C - H, CH_3 - Br\overset{*}{C}H_2 - C \equiv C - H$

48.



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